



Puppet Data-Centre Automation Solution

Part-2

Managing data centres comprising hundreds of systems is tough. When these machines run on *NIX (Linux and other UNIX-like) OSs, the job could get harder. Fortunately, configuration-management systems help to bring things under control by automating common configuration tasks, and hence making systems manageable from a central location. This article focuses on using a prominent FOSS configuration-management system named Puppet, for user and group management.

User and group management has been a very difficult task for systems administrators, especially when one user account has to be created across 10 machines, or even a 100. Similarly, disabling or deleting the accounts on 10s or 100s of machines, when employees leave the team or company, is not for the faint-hearted. You would be right in asking, “Why? Doesn’t LDAP solve the problem?” Believe me, I agree with you! The LDAP (Lightweight Directory Access Protocol) expert, Varad Gupta, has written some beautiful articles on user management in the past few issues of *LINUX For You*. I really enjoyed them and have cut them out for future reference. Then why are we still trying out Puppet for user/group management? I feel, after struggling with LDAP for about two years, that:

- I. Puppet has made the task much simpler for me.
- II. Communications are encrypted by default, without me having to set up SSL separately, which makes it very secure to set even passwords.
- III. The configurations on the client end are very minimal, or nil, compared to LDAP.

In our last session, we had learnt how to set up a basic Puppet server and client architecture.

Today, we will be looking at how to create users and groups, and a little bit of file management in association with user creation.

Groups

We need to create groups first, because a user, when added, needs to be part of a group. Always, management of privileges by a group is better than individual user management. Put all the users into a group, and give the group privileges, controlling all the user accounts in it. This is very useful, even when giving *sudo* privileges. We will not be going to that detail in this example, but will be in an upcoming series.

Enter the following lines in */etc/puppet/manifests/site.pp*. Also, do note that the comma at the end of a line is significant; so don’t omit it.

```
group
{
  "luffyforum":
    ensure => present,
    gid => 1001
}
```

Make sure that a group with the ID 1001 does not already exist.